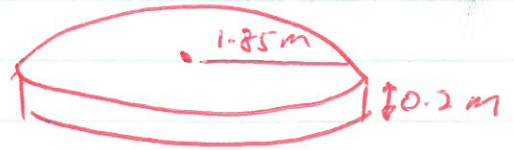
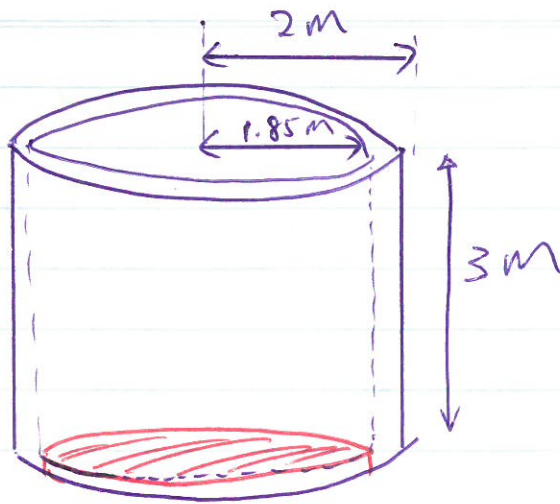


7.



Base

Body

a) volume of concrete required = volume of body + volume of base
 volume of cylindrical body = volume of whole cylinder - vol of hollow

$$= \pi (2)^2 (3) - \pi (1.85)^2 (3)$$

$$= 37.6991 - 32.2563$$

$$= 5.4428 \text{ m}^3$$

$$\approx 5.44 \text{ m}^3$$

volume of base

$$= \pi (1.85)^2 (0.2)$$

$$= 2.1504 \text{ m}^3$$

$$\approx 2.15 \text{ m}^3$$

Total volume of concrete required = $2.15 + 5.44 = 7.59 \text{ m}^3$

b) capacity = volume of hollow - vol of base

$$= \pi (1.85)^2 (3) - \pi (1.85)^2 (0.2)$$

$$= 30.10588 \text{ m}^3$$

$$\approx 30106 \text{ L}$$

c) Area of base = $\pi (2 - 0.15)^2 = 10.7521 \text{ m}^2 \approx 10.75 \text{ m}^2$
 Area of internal wall = $2\pi (1.85)(2.8) = 32.5468 \text{ m}^2 \approx 32.55 \text{ m}^2$
 Total area = 43.30 m^2
 It will require $\frac{43.30}{0.75} = 57.732$ to cover the internal surface
 As it requires 2 coat, the cost is therefore
 $57.73 \times 2 \times \$75 = \8659.80

$$a) \quad c = \begin{matrix} & \text{USB 1} & \text{USB 2} \\ \$ & [0.80 & 0.95] \end{matrix}$$

$$b) \quad T = CP = [0.80 \ 0.95] \begin{bmatrix} 15 & 25 & 35 \\ 20 & 30 & 40 \end{bmatrix} \\ = [31 \ 48.5 \ 66]$$

T is the total monthly production costs of different size of USB.
The cost of producing

1 GB is \$ 31000

2 GB is \$ 48500

5 GB is \$ 66000

$$c) \quad R = \begin{matrix} & \text{USB 1} & \text{USB 2} \\ \$ & [1.50 & 2.10] \end{matrix}$$

$$d) \quad 0.8 \begin{bmatrix} 15 & 25 & 35 \\ 20 & 30 & 40 \end{bmatrix} = \begin{matrix} \text{USB 1} & \text{USB 2} \\ \begin{bmatrix} 12 & 20 & 28 \\ 16 & 24 & 32 \end{bmatrix} \end{matrix}$$

$$e) \quad [1.50 \ 2.10] \begin{bmatrix} 12 & 20 & 28 \\ 16 & 24 & 32 \end{bmatrix} - [31 \ 48.5 \ 66] \\ = [20.6 \ 31.9 \ 43.2]$$

monthly profit:

1 GB = \$ 20600

2 GB = \$ 31900

5 GB = \$ 43200